

Some Continuous Probability Distributions

February 17, 2026

Continuous Uniform (Rectangular) Distribution

- A random variable X is said to have a **uniform** distribution on the interval $[a, b]$, $-\infty < a < b < \infty$, if its **PDF** is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

We will write $X \sim U[a, b]$ if X has a uniform distribution on $[a, b]$.

- The endpoint a or b or both may be excluded. Clearly,

$$\int_{-\infty}^{\infty} f(x) dx = 1,$$

so that (1) indeed defines a PDF.

Continuous Uniform (Rectangular) Distribution (Contd.)

- The **CDF** of X is given by

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x < b, \\ 1, & x \geq b. \end{cases} \quad (2)$$

- Mean:** $E(X) = \frac{a+b}{2}$.
- k^{th} Moment:** $E(X^k) = \frac{b^{k+1}-a^{k+1}}{(k+1)(b-a)}$, $k > 0$ is an integer.
- Variance:** $Var(X) = \frac{(b-a)^2}{12}$.
- MGF:** $M(t) = \frac{e^{tb}-e^{ta}}{t(b-a)}$, $t \neq 0$.

Continuous Uniform (Rectangular) Distribution (Contd.)

Example 1: Let X have a PDF given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & 0 < x < \infty, \lambda > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-\lambda x}, & x > 0. \end{cases}$$

Let $Y = F(X) = 1 - e^{-\lambda x}$. The PDF of Y is given by

$$f(y) = \lambda(1 - y) \left| \frac{dx}{dy} \right| = 1, \quad 0 \leq y \leq 1.$$

Continuous Uniform (Rectangular) Distribution (Contd.)

Let us define $f(y) = 1$ at $y = 1$. Then we see that Y has PDF

$$f(y) = \begin{cases} 1, & 0 \leq y \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

which is the $U[0, 1]$ distribution.

(This is not a mere coincidence, see Theorem 1.)

Continuous Uniform (Rectangular) Distribution (Contd.)

Theorem 1 (Probability Integral Transformation)

Let X be a random variable with a continuous CDF F . Then $F(X)$ has the uniform distribution on $[0, 1]$.

Proof. We have the CDF $F_X(x) = P(X \leq x)$, which is strictly increasing and maps X to values in $[0, 1]$. Let $Y = F(X)$. Now Y will be distributed on $U[0, 1]$ if its CDF is given by $F_Y(y) = y$, $0 \leq y \leq 1$. Using the given information we have

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{F(X) \leq y\} \\ &= P\{X \leq F^{-1}(y)\} = F_X(F^{-1}(y)) \\ &= y. \end{aligned}$$

Thus $Y = F(X)$ is uniformly distributed on $U[0, 1]$.

Continuous Uniform (Rectangular) Distribution (Contd.)

Theorem 2

Let F be any CDF, and let X be a $U[0, 1]$ random variable. Then there exists a function h such that $h(X)$ has CDF F , i.e.,

$$P\{h(X) \leq x\} = F(x) \quad \forall \quad x \in (-\infty, \infty). \quad (3)$$

Proof. If F is the CDF of a discrete random variable Y , let

$$P\{Y = y_k\} = p_k, \quad k = 1, 2, \dots$$

Define h as follows:

$$h(x) = \begin{cases} y_1, & 0 \leq x < p_1, \\ y_2, & p_1 \leq x < p_1 + p_2, \\ \vdots & \vdots \end{cases}$$

Continuous Uniform (Rectangular) Distribution (Contd.)

Then

$$P\{h(X) = y_1\} = P\{0 \leq X < p_1\} = p_1,$$

$$P\{h(X) = y_2\} = P\{p_1 \leq X < p_1 + p_2\} = p_2,$$

and, in general,

$$P\{h(X) = y_k\} = p_k, \quad k = 1, 2, \dots$$

Thus $h(X)$ is a discrete random variable with CDF F .

If F is continuous and strictly increasing, F^{-1} is well defined, and we take $h(X) = F^{-1}(X)$. We have

$$P\{h(X) \leq x\} = P\{F^{-1}(X) \leq x\} = P\{X \leq F(x)\} = F(x),$$

as asserted.

Continuous Uniform (Rectangular) Distribution (Contd.)

Theorem 2 is quite useful in generating samples with the help of the uniform distribution. One can use the following steps to generate random samples in R :

- Start by generating a random variable U from the uniform distribution $U[0, 1]$. This means U is uniformly distributed between 0 and 1, with PDF $f(u) = 1$, $0 \leq u \leq 1$.
- Suppose one wants to generate random samples from a distribution with PDF $f_X(x)$. First, find the CDF $F(x)$ of X .
- Find the inverse of the CDF, $F_X^{-1}(u)$, such that $F_X^{-1}(u)$ returns the value of x corresponding to a given value u from $U[0, 1]$ distribution. Basically, this function gives the value of X for a given cumulative probability u [use of quantile function].
- For each random variable U generated from $U[0, 1]$, apply the inverse CDF to get the corresponding random sample $X = F_X^{-1}(U)$.

Continuous Uniform (Rectangular) Distribution (Contd.)

Example 2: Let F be the CDF defined by

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x} & x > 0. \end{cases}$$

Then the inverse to $y = 1 - e^{-x}$, $x > 0$, is $x = -\log(1 - y)$, $0 < y < 1$.

Thus

$$h(y) = -\log(1 - y),$$

and $-\log(1 - X)$ has the required distribution, where X is a $U[0, 1]$ random variable.

Continuous Uniform (Rectangular) Distribution (Contd.)

R Commands:

- `dunif(x, min, max)` gives the PDF of the uniform distribution.
- `punif(q, min, max)` computes the CDF.
- `qunif(p, min, max)` computes the quantile function for a given cumulative probability ' p '.
- `runif(n, min, max)` generates random samples from the uniform distribution.

Gamma Distribution

- A random variable X with PDF

$$f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & 0 < x < \infty \\ 0, & x \leq 0 \end{cases} \quad (1)$$

is said to have a **gamma distribution** with **shape** and **scale** parameters α and β respectively. We will write $X \sim G(\alpha, \beta)$.

- The CDF of X is given by

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^x y^{\alpha-1} e^{-y/\beta} dy, & y > 0. \end{cases} \quad (2)$$

- The PDF plots for different parameter configurations are given in [Figures 1-4](#).

Gamma Distribution (Contd.)

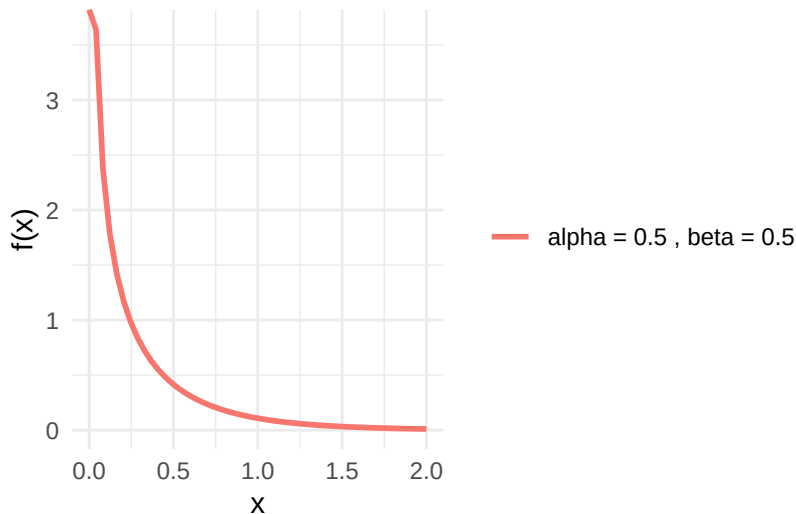


Figure 1: PDF of Gamma Distribution

Gamma Distribution (Contd.)

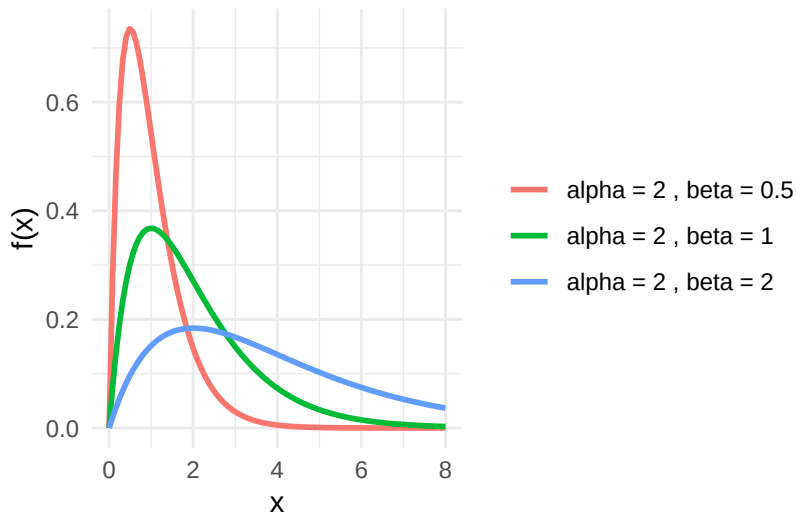


Figure 2: PDF of Gamma Distribution

Gamma Distribution (Contd.)

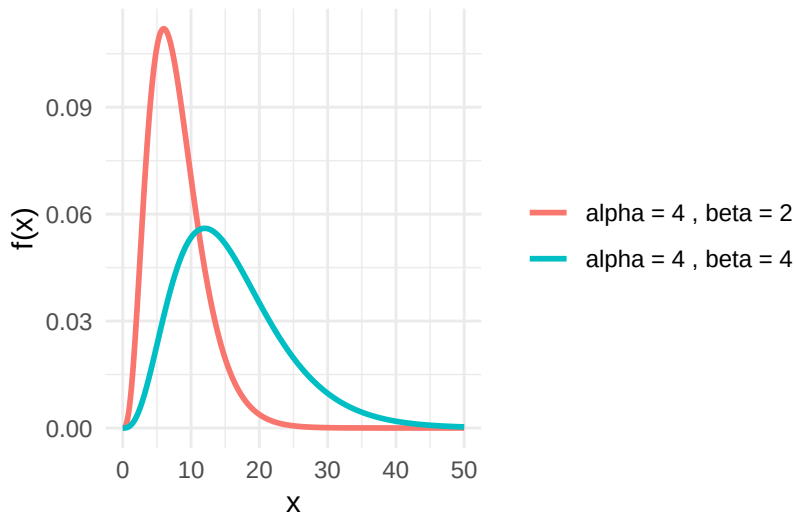


Figure 3: PDF of Gamma Distribution

Gamma Distribution (Contd.)

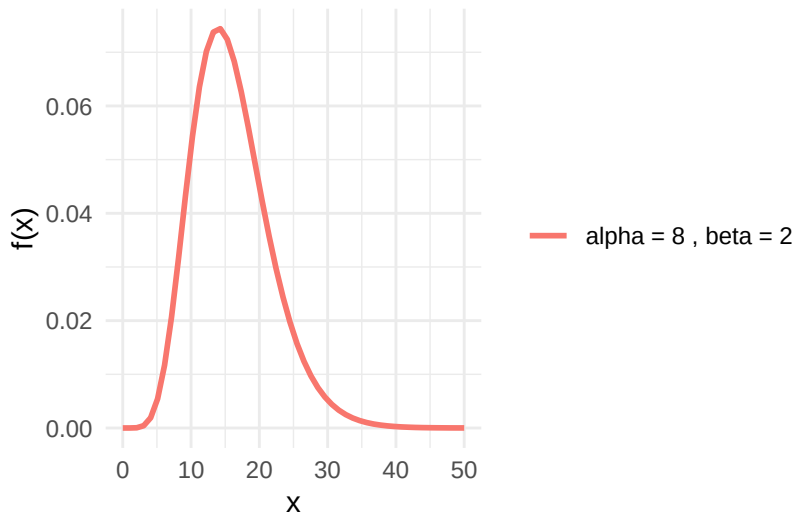


Figure 4: PDF of Gamma Distribution

Gamma Distribution (Contd.)

The **MGF** of X can be calculated as follows:

$$\begin{aligned} M(t) &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{x(t-1/\beta)} dx \\ &= \left(\frac{1}{1-t\beta} \right)^\alpha \int_0^\infty \frac{y^{\alpha-1} e^{-y}}{\Gamma(\alpha)} dy, \quad [\text{substituting } x(t-1/\beta) = -y] \\ &= (1-t\beta)^{-\alpha}, \quad t < 1/\beta. \end{aligned} \tag{3}$$

Gamma Distribution (Contd.)

Using MGF of X (Equation 3), the moments can be obtained as follows:

$$\text{Mean} \equiv E(X) = M'(t)|_{t=0} = \alpha\beta. \quad (4)$$

$$\text{Second Moment} \equiv E(X^2) = M''(t)|_{t=0} = \alpha(\alpha + 1)\beta^2. \quad (5)$$

$$\text{Variance} \equiv \text{Var}(X) = \alpha\beta^2. \quad (6)$$

One can also compute the moments of order n such that $\alpha + n > 0$ directly from the PDF as follows:

$$\begin{aligned} E(X^n) &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty e^{-x/\beta} x^{\alpha+n-1} dx \\ &= \beta^n \frac{\Gamma(\alpha + n)}{\Gamma(\alpha)}. \end{aligned} \quad (7)$$

Gamma Distribution (Contd.)

Applications of Gamma Distribution:

- Gamma distribution can be used to model the distribution of wind speeds over time (In meteorology).
- Gamma distribution can be used to model the failure times of systems/components (In reliability engineering).
- Gamma distribution can be used to model the waiting times in systems with more than one server (In queueing theory).
- Gamma distribution can be used to model the size of insurance claims (In insurance).

Gamma Distribution (Contd.)

R Commands:

- `dgamma(x, shape, scale)` gives the PDF of the gamma distribution.
- `pgamma(q, shape, scale)` computes the CDF.
- `qgamma(p, shape, scale)` computes the quantile function for a given cumulative probability ' p '.
- `rgamma(n, shape, scale)` generates random samples from the gamma distribution.

Exponential Distribution

- The special case when $\alpha = 1$ leads to the **exponential distribution** with **scale** parameter β . The PDF of an exponentially distributed random variable is given by

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

- Clearly, $X \sim \text{Exp}(\beta)$ is equivalent to $X \sim \text{Gamma}(1, \beta)$.

Exponential Distribution (Contd.)

Time between two customer arrivals follow exponential distribution.

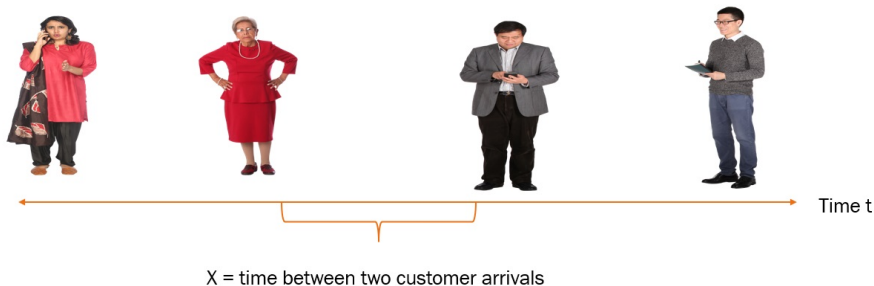


Figure 5: Exponential Distribution (Example)

Exponential Distribution (Contd.)

The expressions of moments and MGF of exponential distribution are given by:

$$\text{Mean} \equiv E(X) = \beta. \quad (9)$$

$$\text{Second Moment} \equiv E(X^2) = 2\beta^2. \quad (10)$$

$$\text{Variance} \equiv \text{Var}(X) = \beta^2. \quad (11)$$

$$E(X^n) = n!\beta^n. \quad (12)$$

$$\text{MGF} \equiv M(t) = (1 - t\beta)^{-1} \text{ for } t < \beta^{-1}. \quad (13)$$

Some Important Properties

Theorem 1

Let $X_1, X_2, \dots, X_n$ be independent random variables such that $X_j \sim \text{Gamma}(\alpha_j, \beta)$, $j = 1, 2, \dots, n$. Then $S_n = \sum_{k=1}^n X_k$ is a $\text{Gamma}(\sum_{j=1}^n \alpha_j, \beta)$ random variable.

Corollary 1. Let $X_1, X_2, \dots, X_n$ be iid random variables, each with an exponential distribution with parameter β . Then S_n is a $\text{Gamma}(n, \beta)$ random variable.

Some Important Properties (Contd.)

Theorem 2

Let $X \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$ be independent random variables. Then $X + Y$ and X/Y are independent.

Corollary 2. Let $X \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$ be independent random variables. Then $X + Y$ and $X/(X + Y)$ are independent.

Theorem 3

Let X and Y be two nondegenerate random variables that take only positive values. Suppose that $U = X + Y$ and $V = X/Y$ are independent. Then X and Y have gamma distribution with the same parameter β .

Some Important Properties (Contd.)

Theorem 4 (Memoryless Property)

Let $X \sim \text{Exp}(\beta)$. Then the random variable X has "no memory", that is,

$$P\{X > r + s | X > s\} = P\{X > r\} \quad (14)$$

for any two positive real numbers r and s .

Theorem 5

If X and Y are independent exponential random variables with parameter β , then $Z = X/(X + Y)$ has a $U(0, 1)$ distribution.

R Commands for Exponential Distribution

- `dexp(x, rate)` gives the PDF of the exponential distribution.
- `pexp(q, rate)` computes the CDF.
- `qexp(p, rate)` computes the quantile function for a given cumulative probability ' p '.
- `rexp(n, rate)` generates random samples from the exponential distribution.

Beta Distribution

- A random variable X with PDF

$$f(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, & 0 < x < 1, \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

is said to have a **beta distribution** with **shape** parameters $\alpha > 0$ and $\beta > 0$ respectively. We will write $X \sim B(\alpha, \beta)$.

- The CDF of X is given by

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{B(\alpha,\beta)} \int_0^x y^{\alpha-1}(1-y)^{\beta-1} dy, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases} \quad (2)$$

Beta Distribution (Contd.)

If n is a positive number, then

$$\begin{aligned} E(X^n) &= \frac{1}{B(\alpha, \beta)} \int_0^1 x^{n+\alpha-1} (1-x)^{\beta-1} dx \\ &= \frac{B(n+\alpha, \beta)}{B(\alpha, \beta)} \\ &= \frac{\Gamma(n+\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(n+\alpha+\beta)}. \end{aligned} \tag{3}$$

In particular, **Mean** $= E(X) = \frac{\alpha}{\alpha+\beta}$, **Variance** $= \text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Beta Distribution (Contd.)

The **MGF** of X can be obtained as follows:

$$M(t) = \frac{1}{B(\alpha, \beta)} \int_0^1 e^{tx} x^{\alpha-1} (1-x)^{\beta-1} dx. \quad (4)$$

Since moments of all order exist, and $E|X|^j < 1$ for all j , we have

$$\begin{aligned} M(t) &= \sum_{j=0}^{\infty} \frac{t^j}{j!} E(X^j) \\ &= \sum_{j=0}^{\infty} \frac{t^j}{\Gamma(j+1)} \frac{\Gamma(\alpha+j)\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+j)\Gamma(\alpha)}. \end{aligned} \quad (5)$$

Beta Distribution (Contd.)

Important Remarks:

- ① When $\alpha = \beta = 1$ we get the uniform distribution on $(0,1)$.
- ② If X is a beta random variable with parameters α and β , then $1 - X$ is a beta random variable with parameters β and α .
- ③ Let X and Y be independent $\text{Gamma}(\alpha_1, \beta)$ and $\text{Gamma}(\alpha_2, \beta)$ random variables respectively. Then $X/(X + Y)$ is a $B(\alpha_1, \alpha_2)$ random variable.

Cauchy Distribution

A random variable X is said to have a **Cauchy Distribution** with parameters μ and θ if its PDF is given by

$$f(x) = \frac{\mu}{\pi} \frac{1}{\mu^2 + (x - \theta)^2}, \quad -\infty < x < \infty, \quad \mu > 0. \quad (6)$$

We will write $X \sim C(\mu, \theta)$ for a Cauchy random variable with PDF (6).

Cauchy Distribution (Contd.)

Theorem 1

Let X be a Cauchy random variable with parameters μ and θ . The moments of order < 1 exist, but the moments of order ≥ 1 do not exist for the random variable X .

Proof. It suffices to consider the PDF, $f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$, $-\infty < x < \infty$.

$$\begin{aligned} E|X|^\alpha &= \frac{2}{\pi} \int_0^\infty \frac{x^\alpha}{1+x^2} dx, \\ &= \frac{1}{\pi} \int_0^1 z^{(1-\alpha)/2-1} (1-z)^{(\alpha+1)/2-1} dz \quad [z = 1/(1+x^2)], \end{aligned}$$

which converges for $\alpha < 1$ and diverges for $\alpha \geq 1$.

Cauchy Distribution (Contd.)

Remark 1. It follows from Theorem 1 that the MGF of a Cauchy random variable does not exist.

Remark 2. We note, however, that the characteristic function of $X \sim C(\mu, 0)$ is given by

$$\phi(t) = e^{-\mu|t|}.$$

Theorem 2

Let X be a Cauchy random variable with parameters μ and θ . The moments of order < 1 exist, but the moments of order ≥ 1 do not exist for the random variable X .

Cauchy Distribution (Contd.)

Theorem 3

Let $X \sim C(\mu_1, \theta_1)$ and $Y \sim C(\mu_2, \theta_2)$ be independent random variables. Then $X + Y$ is a $C(\mu_1 + \mu_2, \theta_1 + \theta_2)$ random variable.

Proof. Let $\mu_1 = \mu_2 = 1$ and $\theta_1 = \theta_2 = 0$ (without loss of generality), i.e., X and Y have the common PDF

$$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

The proof in the general case follows along the same lines. Using the uniqueness property of the characteristic function, one can easily establish that the PDF of $Z = X + Y$ is given by

$$f(z) = \frac{1}{\pi} \frac{2}{z^2 + 2^2}, \quad -\infty < z < \infty.$$

Cauchy Distribution (Contd.)

It follows that if X and Y are iid $C(1, 0)$ random variables, then $X + Y$ is a $C(2, 0)$ random variable.

Corollary. Let $X_1, X_2, \dots, X_n$ be independent Cauchy random variables, $X_k \sim C(\mu_k, \theta_k), k = 1, 2, \dots, n$. Then $S_n = \sum_{k=1}^n X_k$ is a $C\left(\sum_{k=1}^n \mu_k, \sum_{k=1}^n \theta_k\right)$ random variable.

Normal Distribution (Gaussian Law)

- A random variable X is said to have a **Normal Distribution** with parameters μ ($-\infty < \mu < \infty$) and σ (> 0) if its PDF is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty. \quad (7)$$

We will write $X \sim N(\mu, \sigma^2)$.

- $Z = (X - \mu)/\sigma$ is known as a **standard normal variate** and its PDF is given by

$$\psi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty. \quad (8)$$

- Clearly, $Z \sim N(0, 1)$. The CDF of Z is given by

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-t^2/2} dt. \quad (9)$$

Normal Distribution (Contd.)

The **MGF** of $X \sim N(\mu, \sigma^2)$ is given by

$$\begin{aligned} M(t) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-(x-\mu)^2/2\sigma^2} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{-x^2}{2\sigma^2} + \frac{x(\mu + t\sigma^2)}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right) dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[\frac{-(x - \mu - t\sigma^2)^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}\right] dx \\ &= \frac{e^{\mu t + t^2 \sigma^2 / 2}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[\frac{-(x - \mu - t\sigma^2)^2}{2\sigma^2}\right] dx \\ &= e^{\mu t + t^2 \sigma^2 / 2} \end{aligned} \tag{10}$$

for all real values of t .

Normal Distribution (Contd.)

- The MGF of a standard normal variate Z is given by

$$M(t) = e^{t^2/2}.$$

- Moments of all order exist and may be computed from the MGF.

$$\text{Mean} = E(X) = M'(t)|_{t=0} = \mu.$$

$$\text{Second Moment} = E(X^2) = M''(t)|_{t=0} = \sigma^2 + \mu^2.$$

$$\text{Variance} = \text{Var}(X) = \sigma^2.$$

Normal Distribution (Contd.)

Clearly, the central moments of odd order are all zero.

$$\begin{aligned}\mu_{2n+1} &= E(X - \mu)^{2n+1} = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^{2n+1} e^{-(x-\mu)^2/2\sigma^2} dx \\&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z)^{2n+1} e^{-z^2/2} dz \quad [z = (x - \mu)/\sigma] \\&= \frac{\sigma^{2n+1}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n+1} e^{-z^2/2} dz \\&= 0,\end{aligned}$$

since the integrand is an odd function of z .

Normal Distribution (Contd.)

The central moments of even order are as follows:

$$\begin{aligned}\mu_{2n} &= E(X - \mu)^{2n} = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^{2n} e^{-(x-\mu)^2/2\sigma^2} dx \\&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z)^{2n} e^{-z^2/2} dz \quad [z = (x - \mu)/\sigma] \\&= \frac{\sigma^{2n}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n} e^{-z^2/2} dz \\&= \frac{2\sigma^{2n}}{\sqrt{2\pi}} \int_0^{\infty} z^{2n} e^{-z^2/2} dz \quad (\text{integrand is an even function of } z)\end{aligned}$$

Normal Distribution (Contd.)

$$\begin{aligned}\Rightarrow \mu_{2n} &= E(X - \mu)^{2n} = \frac{2\sigma^{2n}}{\sqrt{2\pi}} \int_0^{\infty} (2t)^n \frac{e^{-t}}{\sqrt{2t}} dt \quad [t = z^2/2] \\&= \frac{2^n \sigma^{2n}}{\sqrt{\pi}} \int_0^{\infty} (t)^{n+1/2-1} e^{-t} dt \\&= \frac{2^n \sigma^{2n}}{\sqrt{\pi}} \Gamma(n + 1/2) \\&= [(2n-1)(2n-3)\dots 3.1] \sigma^{2n}.\end{aligned}$$

Normal Distribution (Contd.)

In particular,

$$\mu_1 = 0, \mu_2 = \sigma^2, \mu_3 = 0, \mu_4 = 3\sigma^4.$$

$$\text{Coefficient of Skewness} = \beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.$$

$$\text{Coefficient of Kurtosis} = \beta_2 = \frac{\mu_4}{\mu_2^2} = 3.$$

Remark 2. The normal distribution is one of the most important distributions in probability and statistics, and for this reason the standard normal distribution is available in tabular form.

Normal Distribution (Contd.)

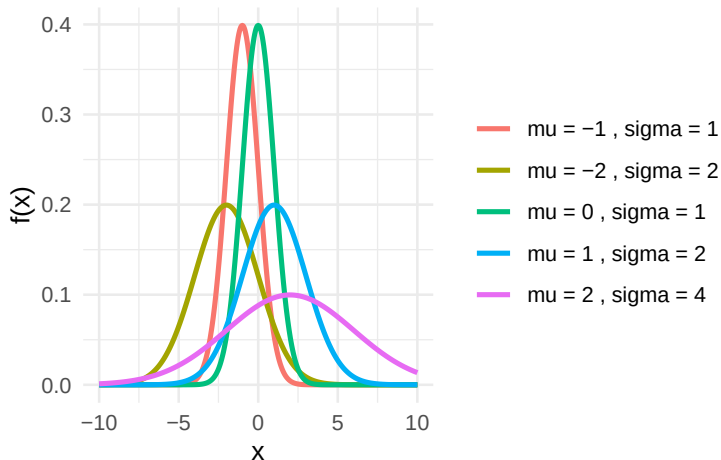


Figure 1: PDF of Normal Distribution

Normal Distribution (Contd.)

R Commands:

- `dnorm(x, mean, sd)` gives the PDF of the normal distribution.
- `pnorm(q, mean, sd)` computes the CDF.
- `qnorm(p, mean, sd)` computes the quantile function for a given cumulative probability ' p '.
- `rnorm(n, mean, sd)` generates random samples from the normal distribution.

Normal Distribution (Contd.)

Area under the standard normal curve:

- $P(-1 \leq Z \leq 1) = 0.6826$ (one standard deviation of the mean).
- $P(-2 \leq Z \leq 2) = 0.9544$ (two standard deviations of the mean).
- $P(-3 \leq Z \leq 3) = 0.9973$ (three standard deviations of the mean).

Thus practically all the distribution is concentrated within three standard deviations of the mean.